Change of basis matrix

Prop.: $A=\{\underline{v}_1,\cdots,\underline{v}_n\}$, $A'=\{\underline{w}_1,\cdots,\underline{w}_n\}$

$$\underline{v}_j=\sum_{i=1}^{n}\alpha_{ij}\underline{w}_i$$

$$\underbrace{\ell_{\phi A'A}[\underline{v}]_A=[\underline{v}]_{A'}}_{(*)},\ \textcircled{1}\in\#^n,\ (\underline{v}\in V)$$

Proof:

① Check (*) holds for $\underline{v}=\underline{v}_j$; (☆! ①→指定任意特值)

LHS(左): $\ell_{\phi A'A}[\underline{v}_j]_A$

$=\ell_{\phi A'A}[0\underline{v}_1+\cdots+1\underline{v}_j+\cdots+0\underline{v}_n]_A$

$=\ell_{\phi A'A}\begin{pmatrix}0\\\vdots\\0\end{pmatrix}\to j^{th}$ entry

$=j^{th}$ col. of $\ell_{\phi A'A}$ (⊗)

$=\begin{pmatrix}\alpha_{1j}\\\vdots\\\alpha_{nj}\end{pmatrix}$

RHS(右): $[\underline{v}_j]_{A'}$ (⊗)

$\overset{(**)}{\underset{①}{=}}[\sum_{i=1}^{n}\alpha_{ij}\underline{w}_i]_{A'}$

$=$ (=LHS)

(☆! ← (② → Generalize ①)

② $\forall \underline{v} = k_1 \underline{v_1} + \cdots + k_n \underline{v_n}$

$$\ell_{\phi A'A}[\underline{v}]_A = \ell_{\phi A'A}[\underline{\quad}]_A$$

($[\ ]_A : V \to \mathbb{F}^n$ is isom, ⇒) $= \ell_{\phi A'A}\left(k_1[\underline{v_1}]_A + \cdots + k_n[\underline{v_n}]_A\right)$

(Shown in MAT2040 ⇒) $= k_1 \ell_{\phi A'A}[\underline{v_1}]_A + \cdots + k_n \cdots$

(Step ① ⇒) $= k_1[\underline{v_1}]_{A'} + \cdots + k_n \cdots$

⇒ $= [k_1 \underline{v_1} + \cdots + k_n \underline{v_n}]_{A'}$

Matrix Representation of Linear Transformation

Def.: Have known: $\underline{v} \in V$, $[\underline{v}]_A \in \mathbb{F}^n$

WTS: $T: V \to W$, $T_{BA} \in M_{m\times n}(\mathbb{F})$

$T: V \to W$

dim n m

basis $A = \{\underline{v_1}, \cdots, \underline{v_n}\}$ $B = \{\underline{w_1}, \cdots, \underline{w_m}\}$

$(T)_{BA} := (\alpha_{ij}) \in M_{m\times n}(\mathbb{F})$ ∈ 存

$T(\underline{v_j}) = \sum_{i=1}^{m} \alpha_{ij} \underline{w_i}$

E.g.: $T: P_3(\mathbb{R}) \to P_1(\mathbb{R})$,

$T(p) = p''$

dim: 4 → 2

basis: $A = \{x^2, 1, x^3, x\}$ → $B = \{x, 1\}$

$T(x^2) = (x^2)'' = (2x)' = 2 = 0 \cdot x + 2 \cdot 1$

$T(x) = 0x + 0 \cdot 1$

$T(x^3) = 6x + 0 \cdot 1$

$T(1) = 0x + 0 \cdot 1$

$\Downarrow$

$T_{BA} = \begin{pmatrix} 0 & 0 & 6 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix} \in M_{2\times 4}(\mathbb{R})$

(注意：方向、顺序)

Theorem 1: If $T: V_A \to W_B$

$$\Downarrow$$

Then $T_{BA}[\underline{v}]_A = [T(\underline{v})]_B$

(可视为: matrix - vector multiplication)

Proof:

Let $A = \{\underline{v}_1, \cdots, \underline{v}_n\}$ be a basis of V

$B = \{\underline{w}_1, \cdots, \underline{w}_m\}$ —— of W

① Check Thm1 holds for $\underline{v} = \underline{v}_j$

LHS: $T_{BA}[\underline{v}_j]_A$

$= T_{BA}(e_j)$ ②

$=$ j-th col. of T_{BA}

$= \begin{pmatrix} \alpha_{1j} \\ \vdots \\ \alpha_{mj} \end{pmatrix}$

RHS: $[T(\underline{v}_j)]_B$

$(***)$ ① $= [\sum_{i=1}^{m} \alpha_{ij} \underline{w}_i]_B$

② $=$ $\Leftarrow$ LHS

② For general $\underline{v} = k_1\underline{v}_1 + \cdots + k_n\underline{v}_n \in V$?

Ex.: Same proof as in the change of basis matrix prop.

Theorem 2:

$$V \xrightarrow{T} W \xrightarrow{S} U$$

	V	W	U
dim	n	m	p
basis	$A=\{v_1,\cdots,v_n\}$	$B=\{w_1,\cdots,w_m\}$	$C=\{u_1,\cdots,u_p\}$

where T_{BA} : a $(m\times n)$ matrix

S_{CB} : $(p\times m)$

$(S\circ T)_{CA}$: $(p\times n)$

Then $(S\circ T)_{CA} = S_{CB} \cdot T_{BA}$

composition of l. trans. — matrix mult.

(relationship ?)

Proof:

① Compare the jth col. of $(S\circ T)_{\ell_A}$ and $S_{\ell_B}\cdot T_{BA}$

$(S\circ T)_{\ell_A}\cdot e_j$ (***)

$= (\sim\sim)\sim [v_j]_A$

(Theorem 1 ⇒) $= [(S\circ T)(v_j)]_{\ell}$

$= [S(T(v_j))]_{\ell}$

② $((***)) = [S(\sum_{i=1}^{m}\alpha_{ij} w_i)]_{\ell}$

(S is linear ⇒) $= [\sum_{i=1}^{m}\alpha_{ij}(S w_i)]_{\ell}$

(****) $= [\sim\sim(\sum_{k=1}^{p}\beta_{ki} u_k)]_{\ell}$

② $= [(\sum_{k=1}^{p})(\sum_{i=1}^{m}\alpha_{ij}\beta_{ki})u_k]_{\ell}$

② $= \begin{pmatrix}\sum_{i=1}^{m}\alpha_{ij}\beta_{1i}\\ \vdots \\ \sum_{i=1}^{m}\alpha_{ij}\beta_{pi}\end{pmatrix}$

② $= \begin{pmatrix}(S_{\ell_B}\cdot T_{BA})_{1j}\\ \vdots \\ (\sim\sim)_{pj}\end{pmatrix}$ = jth col. of $S_{\ell_B}\cdot T_{BA}$

(****)

② $S(w_j) = \sum_{k=1}^{p}\beta_{ki} u_k$

② $S_{\ell_B} = (\beta_{ki})$

Remark: $[\underline{v}]_{A'}$ & $[\underline{v}]_{A}$ are related by the change of basis matrices $C_{A'A}$ & $C_{AA'}$

? $T: V \to W$,

$[T]_{BA}$ & $[T]_{B'A'}$

Corollary: $(T_{B'A'}) = C_{B'B}\,(T_{BA})\,C_{AA'}$

Proof 1:

let $A = \{\underline{v}_1, \cdots, \underline{v}_n\}$, $A' = \{\underline{v}_1', \cdots$) → bases of V

$B = \{\underline{w}_1, \cdots, \underline{w}_m\}$, $B' = \{\underline{w}_1', \cdots$) → $\cdots W$

① Check 两边相等 — 指定任意特情

by applying $e_j = [\underline{v}_j']_{A'}$

RHS: $C_{B'B}\,(T_{BA})\,C_{AA'} \cdot e_j$

$= C_{A'A}\,[\underline{v}_j']_{A'}$

$= (T_{BA})\,[\underline{v}_j']_{A'}$

[Theorem 1] $= C_{B'B}\,[T(\underline{v}_j')]_B$

$= [T(\underline{v}_j')]_{B'}$

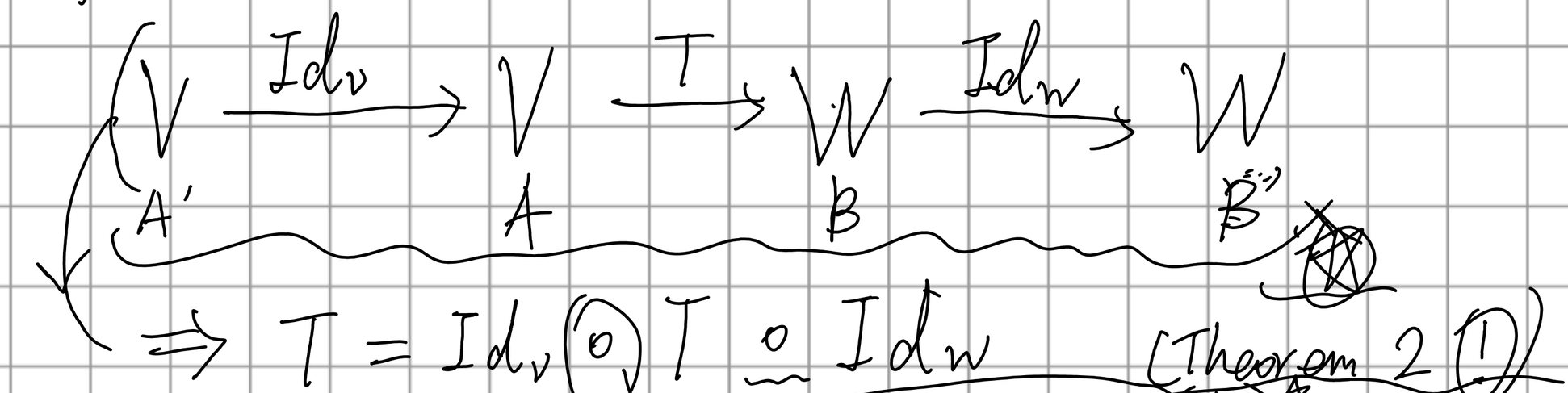
LHS: $T_{B'A'} \cdot e_j$ [先化 e_j 试?] (= !)

$= T_{B'A'} \cdot [\underline{v}_j']_{A'}$

$= [T(\underline{v}_j')]_{B'}$

$=$ RHS

Proof 2:



$\Rightarrow T = Id_V \circ T \circ Id_W$ (Theorem 2 !)

By Theorem 2, $T_{B'A'} = (Id_W)_{B'B} \cdot T_{BA} \cdot (Id_V)_{AA'}$

(In hw ?) $\Rightarrow = \varphi_{B'B} \cdot T_{BA} \cdot \varphi_{A'A}$